Egg and fry production performance of female tilapia related to fluctuating temperature and size variation

Professor Harunur Rashid
Bangladesh Agricultural University

Zahid Parvez Sukhan
Chonnam National University

Egg and Fry Production Performance of Female Tilapia Related to Fluctuating Temperature and Size Variation

Md. Shahadat Hossan[1], Suparna Biswas Ulka[1], Md. Abdul Motin[1],
Md. Abdul Kadir Tarafder[2], Zahid Parvez Sukhan[2], and Harunur Rashid[1]*

[1]Department of Fisheries Management, Bangladesh Agricultural University, Mymensingh 2002, Bangladesh
[2]Sharnalata Agro-Fisheries Ltd., Radhakanai, Fulbaria, Mymensingh, Bangladesh

e-mail: rashid@bau.edu.bd

Abstract

Female Nile tilapia, *Oreochromis niloticus* of six body weight (200, 250, 300, 350, 400 and 450 g) were studied to estimate egg and fry production performance in fluctuating temperature and size of fish at Sharnalata Agro-fisheries Ltd., Mymensingh, Bangladesh. Water temperature was 31-36°C and 29±0.5°C in the broodstock pond and incubator, respectively during experimental period. Histological observation of ovary showed that water temperature has significant effect on egg production ($P < 0.01$). Egg production decreased with increased water temperature and relative fecundity decreased significantly with the increasing body weight of female ($P < 0.01$). Maximum number of eggs, 1342±10.54 and 1377.33±48.27 were found at 32°C and 250 g female, respectively. Histological sections of ovaries from 250 g and 300 g females showed higher number of mature eggs compared to other weight categories. Reproductive performance in *O. niloticus* can be improved by selecting 250 to 350 g brood and rearing them near or below 32°C temperature.

INTRODUCTION

Tilapia has been dubbed as the "aquatic chicken" [1] of which the most widely farmed stock is the Nile tilapia (*Oreochromis niloticus*). Word-Fish Center has developed the genetically improved farmed tilapia (GIFT) strain through several generations of selections involving eight different pure breeds of Nile tilapia *O. niloticus* strain and it is the most widely farmed variety and performs 60% better growth and 50% better survival than other commercially available strains of tilapia [2]. The high growth rate, resistant to adverse environmental and management conditions, and low production cost makes tilapia as the most important commercial species in Bangladesh. The attributes that make Nile tilapia so suitable for fish farming are its resistance against harsh conditions, ease of breeding, rapid growth rate, ability to efficiently convert organic and domestic wastes into high quality protein and good taste [3, 4].

Several environmental and physiological factors are practically involved in fish gonadal maturation and spawning, hence facilitate hatchery production [5]. In natural water bodies, tilapia species can show large variations in its reproductive characteristics [6]. In culture situations, tilapias tend to produce more but smaller oocytes than under natural conditions [7]. Major environmental factors involved in cueing reproductive activity are temperature and photo-period [8–11]. Temperature is generally the most variable environmental parameter and also the most controllable in hatchery condition. It has been considered as the most thoroughly investigated environmental factor influencing fish reproduction.

Sexual maturity in tilapia species is a function of age, size, and environmental conditions. *O. mossambicus*, in general, reaches sexual maturity at a smaller size and younger age than *O. niloticus* and *O. aureus*. Body weight of tilapia influences the number of eggs; medium body weights produce higher number of eggs compared to the small and large body weight. The difference in fecundities (seed/ female/spawning season) could be related to differences in size and age of brood used [12] and the relative fecundity in Nile tilapia has been shown to decrease with increased female size and age [13]. Considering all these facts, the present study was conducted to determine the effect of fluctuating temperature and size of brood on egg and fry production performance of *O. niloticus*.

MATERIALS AND METHODS

A. *Experimental fish and site*

Brood stock of GIFT strain of Nile tilapia were collected from Bangladesh Fisheries Research Institute (BFRI) and reared at the Sharnalata Agro Fisheries Ltd., Mymensingh, Bangladesh. Experiments were conducted during January to August, 2010.

B. *Brood rearing in pond*

Tilapia broods were stocked in ponds at a density of 6-7 per m^2, reared for 3 months, and fed twice daily with floating feed (30% protein) at 1.5-2% body weight. 30% pond water was replaced at 15 days interval.

C. *Selection of brood and stocking in hapa*

Based on the body weight, broods were categorized in six group (200, 250, 300, 350, 400 and 450 g) and stocked in separate breeding hapa ($5m^3$) at a density of 6 per m^2 at 1:2 male-female ratio.

D. *Measurement of temperature, collection of egg, counting, and incubation*

Temperature was measured every day, fertilized eggs were collected at every 15 days interval from the mouth of female tilapia, transported to hatchery, then treated with salt solution, counted carefully, and placed into incubator with controlled water flow of recirculating system.

E. Counting of hatchlings

After hatching out, the hatchlings were counted and hatching rate was calculated using the formula, Hatching rate = (No. of hatched fry/No. of fertilized egg)×100.

F. Histological study of ovary

Ovary of each weight category tilapia at very sampling day were collected and fixed in Bouin's fixative for 24 hrs at 4^{O}C. Gonad tissues were then processed using graded series of ethanol, embedded in paraffin, sectioned at 5 µm thickness, staining using Haematoxylene-Eosin stain, and mounting by DPX mounding agent. Ovary section were then examined under microscope (Olympus).

RESULTS AND DISCUSSION

A. Effect of temperature on egg production

There were direct effect of temperature on egg production of tilapia (Table 1). Egg production decreased with increasing water temperature. Maximum eggs (1342±10.54) were found at 32°C and no egg was found when the temperature was ≥ 35°C. The correlation between water temperature and number of eggs produced per female was negatively significant ($P < 0.01$) and egg production per female decreased with the increasing temperature (Fig. 1).

Table 1. Number of eggs produced per female tilapia (~ 250 g weight) at different temperature.

Water temperature (°C)	Number of eggs/female
31.5	218.66±15.50
32.0	1342.00±10.54
33.0	1305.00±8.89
33.5	801.66±22.55
34.0	636.33±39.27
35.0	0

Temperature is considered as the crucial cue in gonadal development in many fish species [10, 11]. Temperature of 21-23°C is the minimum range required for spawning of fish [14]. Temperatures above 20°C trigger the development of secondary sexual characteristics and nest building [7]. Nile tilapia does not lay eggs in water temperature of below 19°C and above 32°C, and the most favorable productive period coincided with a rise in water temperature range from 22 to 27°C [11].

B. Effect of body weight on egg production

Numbers of eggs per female were significantly higher in 250~300 weight tilapia compared with other weight category (Table 2) and higher weight fish tend to produce less number of eggs (Fig. 2).

Table 2. Numbers of eggs produced per female tilapia at different sizes.

Weight of female (g)	Number of eggs/female
~ 200	1201.00±27.62
~ 250	1377.33±48.27
~ 300	1298.66±23.81
~ 350	1216.33±49.37
~ 400	989.67±40.00
~ 450	820.67±9.02

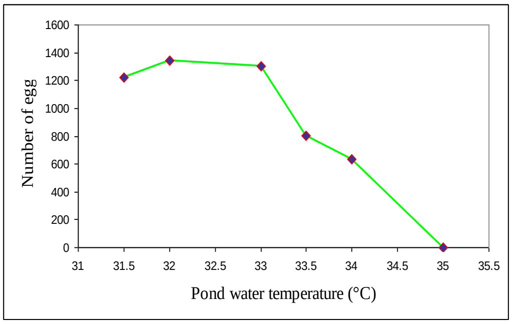

Fig. 1. Relationships between number of egg and water temperature at 250 g body weight of female *Oreochromis niloticus.*

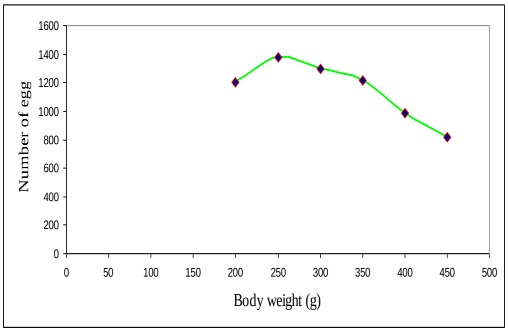

Fig. 2. Relationships between number of egg and body weight of *Oreochromis niloticus* at 32°C temperature.

Number of egg and body weight of female showed inverse relationship [15, 16], similar findings were observed in the present study. The difference in fecundities (seed/female/spawning season) could be related to differences in size and age of brood used [12]. Comparison of reproductive performance of tilapia could be a complicated issue because it is affected by brood size, spawning history, production setting, and the limitation of broodstock selection. Unfortunately, broodstocks used in the present study were selected randomly and largely of unknown age. *Oreochromis* species is a mouth-brooder in nature with small gonads and produce less than 700 eggs [17] and fecundity ranged from 243 to 1847 eggs in all sizes [18]. The present study confirmed that fecundity was significantly related to fish body weight (1377.33±48.27 ~ 820.67±9.02 eggs in 250 g ~ 450 g tilapia).

C. Effect of body weight on stages of ovarian cells

Five different stages of ovarian cells were observed in the histological observation of different weight tilapia and mentioned in Table 3 in detail.

Table 3. Intensity of ovarian cells observed in different body weight tilapia.

Body weight	EP	LP	YVS	PM	ME	Fig. No.
~ 200	++	++	+	+	+++	Fig. 3
~ 250	+	×	+	×	+++	Fig. 4
~ 300			+	×	+++	Fig. 5
~ 350	+++	+++	×	×	+++	Fig. 6
~ 400	+	×	×	×	+	Fig. 7
~ 450	+	×	×	×	+	Fig. 8

EP, early perinucleolar; LP, late perinucleolar, YVS, yolk- vesicle (YVS), PM, pre-maturation; ME, mature egg. +: very few; ++: few; +++: more; ×: none

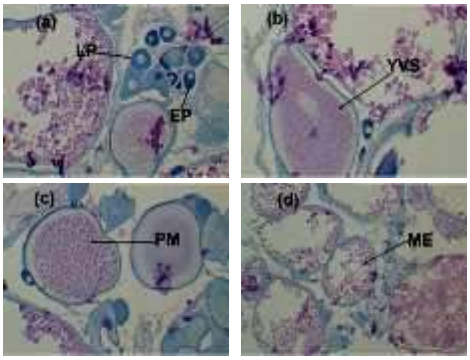

Fig. 3. Haematoxylin-eosin stained sections of tilapia ovary at 200g body weight fish (a-d). (a) EP, early perinucleolar stage; LP, late perinucleolar stage (b) YVS, yolk vesicle stage. (c) PM, pre-maturation stage. (d) ME, mature egg.

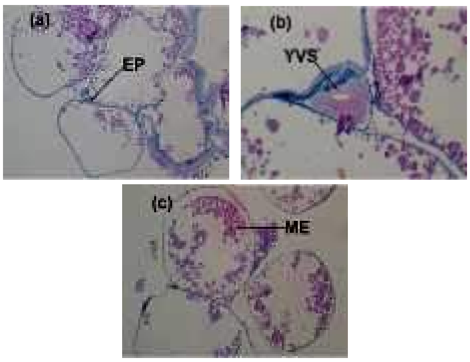

Fig. 4. Haematoxylin-eosin stained sections of tilapia ovary at 250g body weight fish (a-c). (a) EP, early perinucleolar stage; (b) YVS, yolk vesicle stage. (c) ME, mature egg.

250g and 300g female tilapia contained the highest ME compared to others stages with very few YVS and EP oocytes. 350 g weight female contained higher number of early stage (EP, LP) and few numbers of later stage (YVS, ME) oocytes. Besides the highest number of eggs (1377.33±48.27), more mature eggs were evident in the 250 g female. Numbers of mature eggs in histological sections were decreased with the increasing body weight. This results are also correlated with previous histological study that the ovaries of *O. niloticus* contained less number of mature egg with increasing body weight [19].

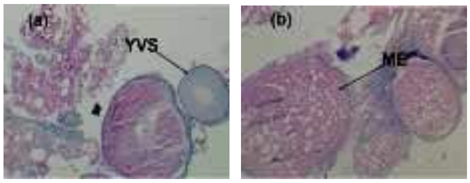

Fig. 5. Haematoxylin-eosin stained sections of tilapia ovary at 300g body weight fish (a-b). (a) YVS, yolk vesicle stage. (b) ME, mature egg.

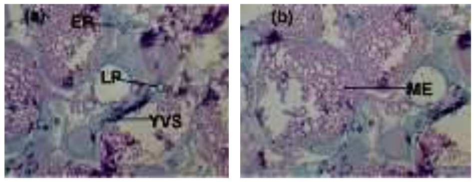

Fig. 6. Haematoxylin-eosin stained sections of tilapia ovary of 350g body weight fish (a-b). (a) EP, early perinucleolar stage; LP, late perinucleolar stage; YVS, yolk vesicle stage. (b) ME, mature egg.

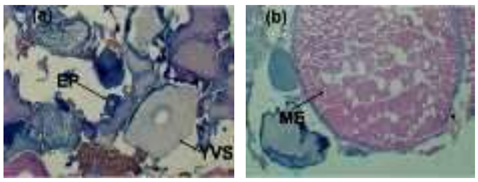

Fig. 7. Haematoxylin-eosin stained sections of tilapia ovary at 400g body weight fish (a-b). (a) EP, early perinucleolar stage ; YVS, yolk vesicle stage . (b) ME, mature egg.

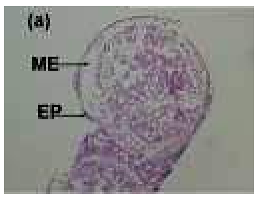

Fig. 8. Haematoxylin-eosin stained sections of tilapia ovary at 450g body weight fish. (a) EP, early perinucleolar stage; ME, mature egg.

CONCLUSION

It can be concluded that temperature and body weight have a significant effect on egg production performance of tilapia. For Nile tilapia broodstock pond, it better to maintain water temperature at 31-32°C. Moreover medium size body weights (250 and 300 g) are good for egg production compared to small and large size fish. For better tilapia hatchery management temperature and body weight of brood should be managed in recommended ways.

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